When a Model Fabricates, It Stops Reading the Prompt: Cheap Claim-Level Hallucination Detection from Attention

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Abstract

Claim-level hallucination detection for long-form LLM output remains hard and expensive: on the MUCH benchmark the strongest published method, Claim-Conditioned Probability (CCP), reaches 0.77 ROC-AUC while costing more compute than generation itself. We show that when an LLM states a false claim, its attention measurably *retrieves less from the prompt* and *leaks more onto the attention sink*, an effect consistent across four models and visible at every depth. GRASP turns this into a detector: three interpretable statistics (grounding, sink leakage, dispersion) from a few label-free-selected heads per layer, plus the token probabilities the model already produced, fed to a small gradient-boosted classifier. On MUCH's human-annotated test split GRASP reaches 0.924 ROC-AUC against CCP's 0.770 (0.943 vs 0.811 against human consensus labels), wins on every model and claim type, and costs 33–38% of generation time on a T4 where CCP costs 192–430% on the same hardware. On RAGTruth, trained with labels from the target benchmark it beats CCP and the pre-trained UHeads of Shelmanov et al. (2025) (0.827 vs 0.623 and 0.680); trained on MUCH plus two RAGTruth task types, it outperforms both on the held-out third task (pooled 0.723 vs 0.623 and 0.680). Trained on a single benchmark and moved to a different task family, it does not: there pre-trained UHeads transfer better. Training-free variants do not beat CCP.

1 Introduction

LLMs state invented facts as fluently as true ones, so a safe deployment needs to know *which* claim to check, not merely that an answer is risky (Min et al., 2023). MUCH (Dentan et al., 2026) makes claim-level detection measurable at scale: 4,873 long-form answers from four open models in four languages, segmented into 20,751 claims, with generated tokens and top-24 probabilities released. Its best baseline, CCP (Fadeeva et al., 2024), filters token uncertainty through a natural-language-inference model at every token. It is accurate (0.772 ROC-AUC) but costs more than generation.

We ask what the model's own attention reveals. Our first finding is mechanistic: across four models, attention during a hallucinated claim is less grounded in the prompt and more often parked on the attention sink (§5). Our second is practical: a small classifier on a handful of such statistics, plus free token probabilities, beats CCP by a wide margin at a fraction of its cost, and with diverse supervision it transfers to unseen task types better than a large pre-trained detector. Our third is a set of explicit negative results: no zero-shot win across benchmark families when trained on one benchmark, and no training-free win.

Contributions. (1) Evidence, consistent across four models and all depths, that hallucinated claims show lower prompt grounding and higher sink leakage. (2) GRASP, an interpretable, cheap, claim-level detector built on it. (3) An evaluation that keeps every choice on training data and reports against GPT-4o and human labels, per model and claim type, across benchmarks, against CCP and UHeads, with runtime measured on the same GPU. (4) An honest map of where it does not work.

Method	Granularity	Signal	Extra passes	La
CCP (Fadeeva et al., 2024)	claim	probabilities + NLI	NLI per token	
SelfCheckGPT (Manakul et al., 2023)	sentence	sample agreement	N samples	
Semantic entropy (Kuhn et al., 2023)	sequence	sample meaning	N samples	
SAPLMA (Azaria and Mitchell, 2023)	sentence	hidden states	none	
Lookback Lens (Chuang et al., 2024)	span	context/self attention ratio	none	
RAUQ (Vazhentsev et al., 2026)	sequence	attention to previous token	none	
UHeads (Shelmanov et al., 2025)	claim	attention + probabilities	transformer head	
GRASP	claim	grounding, sink, dispersion + probs	one replay	yes (free

Table 1: Related detectors.

2 Related work

Probability-based scores such as token likelihood and entropy are cheap but conflate lexical and factual uncertainty; CCP (Fadeeva et al., 2024) removes lexical variation with NLI at a large cost. **Sampling-based** methods such as SelfCheckGPT (Manakul et al., 2023) and semantic entropy (Kuhn et al., 2023) multiply cost and miss consistent errors. **Internal-state probes** predict hallucination from hidden states (Azaria and Mitchell, 2023; Chen et al., 2024) or attention: Lookback Lens (Chuang et al., 2024) uses the ratio of attention to context versus generated text, SinkProbe (Binkowski et al., 2026) the sink mass, and RAUQ (Vazhentsev et al., 2026) label-free head selection at sequence level. Lee et al. (2026) find that simple regularised probes suffice in a paired setting. Closest to us, UHeads (Shelmanov et al., 2025) train a transformer head on attention and probability features for claim-level detection. GRASP differs in using three interpretable statistics from a few label-free-selected heads and a small tree model; we compare to UHeads directly.

3 Method

3.1 Per-token attention signals

The generated tokens are replayed once through the model. For generated token t and a kept head, with attention row a_t over positions $0..t$ (position 0 is the sink, $1..P{-}1$ the prompt), we compute grounding to the prompt G_t, sink leakage S_t and dispersion D_t:

$$G_t = \frac{\sum_{j=1}^{P-1} a_{t,j}}{\sum_{j\geq 1} a_{t,j}}, \qquad S_t = a_{t,0}, \tag{1}$$

$$D_t = \frac{-\sum_{j\geq 1} \tilde{a}_{t,j} \log \tilde{a}_{t,j}}{\log |\{j \geq 1\}|}, \tag{2}$$

where $\tilde{a}$ is the non-sink row renormalised, and the row used is the one that produced t.

3.2 Label-free head selection

Per layer we keep the $k{=}4$ heads whose G varies most across tokens on 200 unlabelled answers from the same model; a head with near-constant G cannot discriminate.

3.3 Claim features and classifier

Layers are pooled into 8 relative-depth bins so models of different depth share one feature space. For each bin, kept head and signal, a claim's content tokens give the mean, max, min and a saliency-weighted mean of the per-(layer, head) z-scored value (384 features). Seven features come from the token probabilities already produced (chosen-token surprisal, entropy, top-1 probability, top-1/top-2 gap) and two from claim length. A gradient-boosted tree classifier (31 leaves, 800 rounds) predicts the label. **GRASP-free** averages, without labels, the per-model rank of mid-depth minimum grounding and of token surprisal and entropy, with signs fixed a priori.

4 Experimental setup

Benchmarks. *MUCH*: all 8 configurations (Llama-3.2-3B, Llama-3.1-8B, Gemma-3-4B, Ministral-8B; temperature 0.7 and 1.0). Test is the 200 human-annotated answers (867 claims); train the other 4,673 (19,884 claims). Following MUCH's evaluator we score against GPT-4o labels on claims where GPT-4o and GPT-4.1 agree, and additionally against the two human annotators. *RAGTruth* (Niu et al., 2024): Mistral-7B-Instruct responses to retrieval-augmented QA, summarisation and data-to-text; claims from MUCH's segmenter, hallucinated if they overlap an annotated span; 445 test responses (17,226 claims). *AkimBio*: **[pending]**.

Protocol. Every modelling choice was made by 5-fold cross-validation on training data, grouped by answer and ranked by the *mean of per-model* ROC-AUCs (pooled ROC can be inflated by recognising the model, as hallucination rates range from 22% to 54%). Choices were frozen to a file before any test split was read; evaluation rules for RAGTruth were written before its features existed. MUCH's test split was read by an earlier version too; we report this.

Baselines. CCP as released by MUCH (δ=24, σ=8, product aggregation), reproduced at 0.775 / 0.643 ROC/PR (published 0.772 / 0.639), and re-run with MUCH's own code on RAGTruth; token surprisal and entropy; UHeads via the authors' pre-trained claim heads, run on identical claims, and retrained on our training data (**[pending]**).

Statistics. 95% CIs from 2,000 bootstrap resamples of answers; differences paired on identical claims.

5 Why it works: hallucinated claims are less grounded

Figure 1 shows the raw signals before any classifier. For hallucinated claims, grounding to the prompt is lower at every depth ($d \approx -0.4$ to -0.5, strongest in the first half of the network), and sink leakage is higher from about 12% depth onward ($d \approx +0.3$ to $+0.4$). Dispersion shows no consistent effect ($|d| < 0.16$). When the model fabricates, it reads less from what it was asked and parks more attention on the sink. A permutation analysis of the trained classifier (Figure 2) agrees: among attention groups, sink leakage at 62–75% depth contributes most, dispersion least. Token probabilities carry the largest single share, and the attention groups are mutually redundant, so each looks small alone. Together they lift cross-validated ROC from 0.771 (probabilities alone) to 0.866.

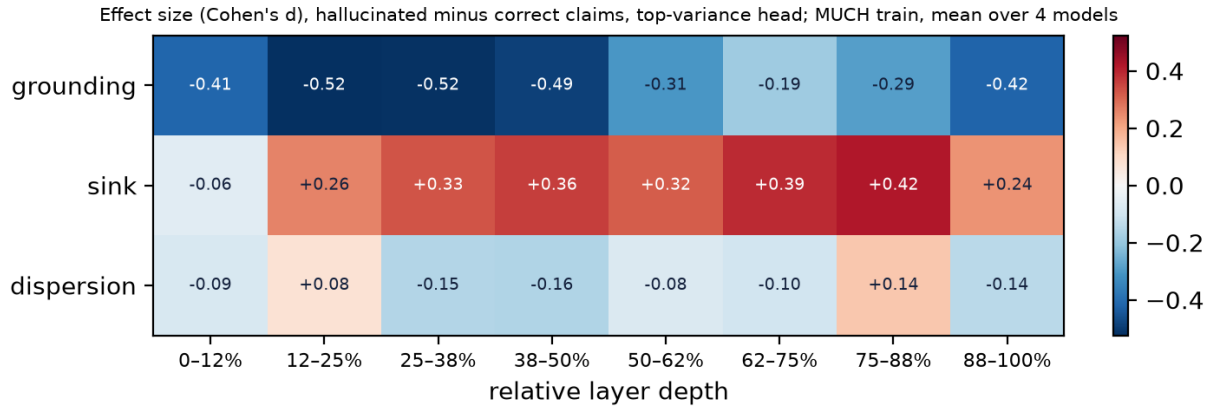


Figure 1: Effect size (Cohen's d) between hallucinated and correct claims for the mean signal over a claim's content tokens, top-variance head, by relative depth; MUCH training claims, averaged over four models. Blue: lower for hallucinated claims.

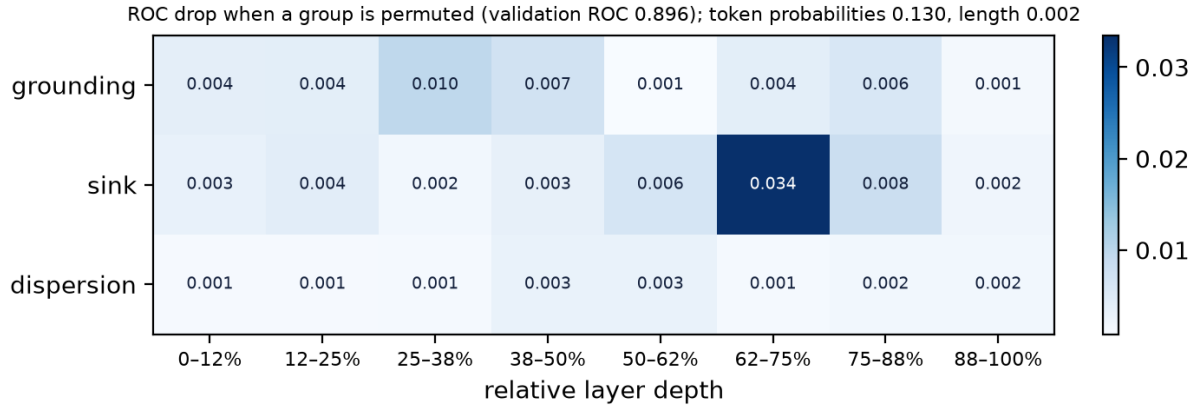


Figure 2: ROC drop when a feature group is permuted, GRASP trained on 80% of MUCH training answers and measured on the other 20%.

6 Results

6.1 MUCH

GRASP beats CCP on every model (Gemma-3-4B 0.917 vs 0.726, Llama-3.1-8B 0.937 vs 0.831, Llama-3.2-3B 0.889 vs 0.767, Ministral-8B 0.897 vs 0.830; all paired CIs exclude zero) and against each human annotator separately (0.928 and 0.939 vs 0.802 and 0.808). Across five runs on different 90% samples of the training answers, test ROC is 0.916 ± 0.003 (human labels 0.936 ± 0.004). At response level (any hallucination in the answer) it reaches 0.911 vs 0.762.

Claim types. GRASP wins on numbers and dates (0.960 vs 0.867), named entities (0.937 vs 0.779) and other claims (0.865 vs 0.659), with the largest gain where CCP is weakest.

Unseen model. Trained on three models and scored on the fourth, GRASP beats CCP on three and trails on Llama-3.2-3B (0.759 vs 0.767).

Method	GPT-4o labels (867 claims) ROC-AUC [95% CI]	PR-AUC	TPR@10%	human consensus (842) ROC-AUC	Δ vs CCP
CCP	0.770 [0.724, 0.817]	0.637	0.537	0.811	–
GRASP-free	0.764 [0.713, 0.815]	0.501	0.354	0.777	−0.034
GRASP	**0.924** [0.896, 0.948]	**0.845**	**0.790**	**0.943**	**+0.132** [+0.091, +0.178]

Table 2: MUCH human-annotated test split. GRASP's ΔROC vs CCP on GPT-4o labels: +0.154 [+0.110, +0.199].

Training data	Method	ROC-AUC [95% CI]	ΔROC vs CCP
–	CCP	0.623 [0.602, 0.644]	–
authors' data	UHead (pre-trained)	0.680 [0.649, 0.708]	+0.057 [+0.029, +0.083]
MUCH only	GRASP + rank normalisation	0.639 [0.617, 0.659]	+0.015 [−0.011, +0.042]
MUCH + other 2 RAGTruth tasks	GRASP (held-out task, pooled)	**0.723** [0.700, 0.746]	**+0.099** [+0.071, +0.129]
RAGTruth train	GRASP	**0.827** [0.803, 0.849]	**+0.203** [+0.177, +0.233]

Table 3: RAGTruth test, claim level (UHead row on its 17,181 covered claims).

6.2 RAGTruth

Table 3 separates three regimes. With RAGTruth's own labels GRASP beats CCP by 0.20 and the pre-trained UHead by 0.15. Trained on MUCH alone it reaches only CCP's level, even after a label-free per-domain rank normalisation selected by MUCH leave-one-model-out (0.808 → 0.819 held-out ROC on MUCH). There the pre-trained UHead, trained on far more varied data, transfers better. With moderately diverse supervision (MUCH plus two RAGTruth task types), GRASP beats both on the held-out task: QA 0.795 (UHead 0.644), summaries 0.737 (0.676), data-to-text 0.679 (0.702, CI of the difference crosses zero); pooled +0.043 over UHead [+0.009, +0.077]. In RAGTruth's retrieval-augmented setting, attention matters more than probabilities: in training cross-validation attention alone gives 0.785, probabilities alone 0.619, both 0.801.

6.3 Comparison with UHeads

On MUCH's Llama-3.1-8B test claims, the authors' pre-trained Llama head scores 0.570, against 0.831 for CCP, 0.883 for GRASP trained on the other three models and 0.937 in-domain. The head is weak on MUCH even on training claims (oriented ROC 0.526) while the same code is sensible on RAGTruth, which points to a mismatch with MUCH's very short claims. A same-data comparison with UHeads retrained on our training splits is **[pending]**.

6.4 Runtime and abstention

On Tesla T4s, GRASP's extraction pass costs 33–38% of measured generation time; MUCH's CCP, timed on the same hardware, costs 235–430% on MUCH's models and 192% on RAGTruth answers. For selective answering, withholding the 30% of claims GRASP rates riskiest leaves 7.2% hallucinated among the rest (CCP: 14.7%; base rate 26.4%).

7 Limitations

GRASP is supervised: its wins need labels from the target setting, or at least diverse related supervision; trained on one benchmark and moved to a different task family it does not beat a large pre-trained head. Training-free variants do not beat CCP at claim or response level. MUCH's test split was read more than once (all choices frozen on training data beforehand), and further reads of RAGTruth test were made for additional analyses. Evaluation covers two benchmarks with released claim labels and five generator models; a third benchmark is **[pending]**. RAGTruth's span labels miss many hallucinations (Yu et al., 2026), depressing absolute scores. CCP and generation were timed on different sample generations (compared per token). RAGTruth responses were replayed through Mistral-7B-Instruct-v0.1, assumed to be the generator version.

8 Conclusion

Hallucinated claims leave a consistent trace in attention: less grounding in the prompt, more leakage onto the sink. Reading that trace from a few heads, together with the probabilities the model already computed, gives a claim-level detector that beats CCP and pre-trained UHeads when trained with relevant labels, at a sixth of CCP's cost, and transfers to unseen task types when the supervision is diverse. Transfer across very different task families without such supervision remains open.

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